



Multiplying Polynomials

To multiply polynomials, use various properties, including the distributive property.

Multiplying Two Monomials

To multiply monomials, rearrange the factors so that the constants are grouped together and variable factors with the same base are grouped together. In other words, just put the constants together, the x 's together, the y 's together, and so on.

EXAMPLE 1

Multiply $-3x^3y^2z^6 \cdot 2x^4y$.

SOLUTION

$$\begin{aligned} -3x^3y^2z^6 \cdot 2x^4y &= -3 \cdot 2 \cdot x^3x^4 \cdot y^2y \cdot z^6 && \text{Commutative and Associative Properties} \\ &= -6x^7y^3z^6 && \text{Product of Powers} \blacksquare \end{aligned}$$

Multiplying a Polynomial by a Monomial

The distributive property states that for any real numbers a , b , and c , $a(b + c) = ab + ac$. The distributive property also applies when the order is changed or when there are more than two terms in the group. For example,

$$(b + c)a = ba + ca \text{ and } a(b + c + d) = ab + ac + ad.$$

EXAMPLE 2

Multiply.

$$5a(2a^2 - 7a - 4)$$

SOLUTION

$$\begin{aligned} 5a(2a^2 - 7a - 4) &= 5a \cdot 2a^2 + 5a \cdot (-7a) + 5a \cdot (-4) \\ &= 10a^3 - 35a^2 - 20a \end{aligned}$$

Distributive Property

Multiply the monomials. ■

Multiplying a Polynomial by a Polynomial

EXAMPLE 3

Multiply and simplify.

A $(3x + 2)(x - 1)$

SOLUTION

$$(3x + 2)(x - 1) = 3x(x - 1) + 2(x - 1)$$

$$= 3x \cdot x + 3x \cdot (-1) + 2 \cdot x + 2 \cdot (-1)$$

$$= 3x^2 - 3x + 2x - 2$$

$$= 3x^2 - x - 2$$

Think of $(x - 1)$ as a single quantity. Apply the distributive property. Multiply $(x - 1)$ by $3x$ and then by 2 .

Apply the distributive property to multiply $3x(x - 1)$. Apply the distributive property to multiply $2(x - 1)$.

Multiply the monomials.

Combine like terms.

B $(x + 3)(x^2 - 2x - 1)$

SOLUTION

$$\begin{aligned}(x + 3)(x^2 - 2x - 1) &= x(x^2 - 2x - 1) + 3(x^2 - 2x - 1) \\&= x \cdot x^2 + x \cdot (-2x) + x \cdot (-1) + 3 \cdot x^2 + 3 \cdot (-2x) + 3 \cdot (-1) \\&= x^3 - 2x^2 - x + 3x^2 - 6x - 3 \\&= x^3 + x^2 - 7x - 3\end{aligned}$$

C $(x^3 + 2x^2 - x)(x^2 - 4x - 3)$

SOLUTION

Use the distributive property to multiply.

$$\begin{aligned}(x^3 + 2x^2 - x)(x^2 - 4x - 3) &= x^3(x^2 - 4x - 3) + 2x^2(x^2 - 4x - 3) - x(x^2 - 4x - 3) \\&= x^3 \cdot x^2 + x^3 \cdot (-4x) + x^3 \cdot (-3) + 2x^2 \cdot x^2 + 2x^2 \cdot (-4x) \\&\quad + 2x^2 \cdot (-3) - x \cdot x^2 - x \cdot (-4x) - x \cdot (-3) \\&= x^5 - 4x^4 - 3x^3 + 2x^4 - 8x^3 - 6x^2 - x^3 + 4x^2 + 3x \\&= x^5 - 2x^4 - 12x^3 - 2x^2 + 3x \quad \blacksquare\end{aligned}$$

Multiplying Two Binomials Using the FOIL Method

When you apply the distributive property to multiply two polynomials, each term in one polynomial is multiplied by each term in the other polynomial.

When both polynomials are binomials, you can use the FOIL method as a way to organize your steps. The letters of the word **FOIL** stand for **F**irst, **O**uter, **I**nnner, and **L**ast.

► **Think About It** The FOIL method combines the distributive property and the commutative property.

EXAMPLE 4

Multiply $(x + 5)(3x - 2)$.

SOLUTION

$$\begin{aligned} (x + 5)(3x - 2) &= \overset{\text{First}}{3x^2} - \overset{\text{Outer}}{2x} + \overset{\text{Inner}}{15x} - \overset{\text{Last}}{10} \\ &= 3x^2 - 2x + 15x - 10 && \text{Identify like terms.} \\ &= 3x^2 + 13x - 10 && \text{Combine like terms.} \end{aligned}$$

Writing Polynomial Models for Areas of Polygons

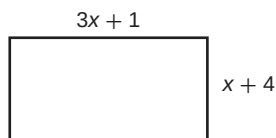
Polynomials can be used to represent areas of plane figures.

EXAMPLE 5

A Write a simplified polynomial to represent the area of the rectangle.

SOLUTION

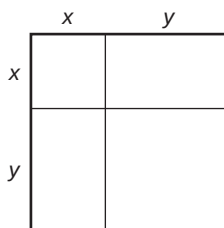
Use the formula for the area of a rectangle.



area = length $\times$ width

$$\begin{aligned} A &= (3x + 1)(x + 4) \\ &= 3x^2 + 12x + x + 4 && \text{FOIL method} \\ &= 3x^2 + 13x + 4 && \text{Combine like terms.} \end{aligned}$$

- B** Use two different methods to write a simplified polynomial that represents the area of the large square.



SOLUTION

Method 1 The area of the large square is the sum of the four areas.

$$x^2 + xy + xy + y^2, \text{ or } x^2 + 2xy + y^2$$

Method 2 The length of each side of the large square is $x + y$. Use the formula for the area of a square.

area = side length $\times$ side length

$$A = (x + y)(x + y)$$

$$= x^2 + xy + xy + y^2 \quad \text{FOIL method}$$

$$= x^2 + 2xy + y^2 \quad \text{Combine like terms. } \blacksquare$$